

Geometric Sequences and Series

ANSWER KEY



Geometric sequences

- 1 Consider the sequence 3, 6, 12, 24, ...
- a** State the common ratio. $r = 2$ **b** Write a formula for the n th term. $a_n = 3 \cdot 2^{n-1}$ **c** Find a_{10} . $a_{10} = 1536$
- 2 In a geometric sequence, $a_2 = 6$ and $a_5 = 48$. Find the common ratio and the first term.
 $r^3 = \frac{48}{6} = 8$, so $r = 2$ and $a_1 = 3$.
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Geometric series

- 3 Find the sum of the first 8 terms of $5 + 10 + 20 + \dots$
 $S_8 = 5 \frac{2^8 - 1}{2 - 1} = 5(255) = 1275$.
- 4 Evaluate each infinite sum, or state that it diverges:
- a** $18 + 6 + 2 + \frac{2}{3} + \dots$ $r = \frac{1}{3}$: $S = \frac{18}{1 - 1/3} = 27$.
- b** $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots$ $r = -\frac{1}{2}$: $S = \frac{1}{1 + 1/2} = \frac{2}{3}$.
- c** $2 + 3 + 4.5 + \dots$ $r = 1.5 > 1$: diverges.
- 5 Write the repeating decimal $0.\overline{7}$ as a fraction using an infinite geometric series.
 $0.7 + 0.07 + \dots$ with $a = \frac{7}{10}$, $r = \frac{1}{10}$: $S = \frac{7/10}{9/10} = \frac{7}{9}$.
- 6 A ball is dropped from 8 m and rebounds to $\frac{3}{4}$ of its previous height on each bounce. Find the total vertical distance it travels (falling and rising) before coming to rest.
Total = $8 + 2\left(8 \cdot \frac{3}{4} + 8 \cdot \left(\frac{3}{4}\right)^2 + \dots\right) = 8 + 2 \frac{6}{1 - 3/4} = 8 + 48 = 56$ m.
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Geometric means and mixed sequences

- 7 Insert two positive geometric means between 4 and 108 (i.e., find b_1, b_2 so that 4, $b_1, b_2, 108$ is geometric).
4 terms total: $r^3 = \frac{108}{4} = 27$, so $r = 3$. Means: 12, 36.
- 8 Determine whether each sequence is arithmetic, geometric, or neither:
- a** 2, 6, 18, 54, ... Geometric ($r = 3$) **b** 5, 8, 11, 14, ... Arithmetic ($d = 3$) **c** 1, 4, 9, 16, ... Neither (differences 3, 5, 7 are not constant, and ratios are not constant)

Applications

9 A car worth 80,000 riyals depreciates by 15% per year.

a Write a formula for its value after n years. $V_n = 80000(0.85)^n$.

b Find its value after 5 years, to the nearest riyal. $V_5 = 80000(0.85)^5 \approx 80000(0.4437) \approx 35,497$ riyals.

10 A chain letter starts with 1 person sending it to 5 people, each of whom sends it to 5 more, and so on. Find the total number of people who have received the letter after 6 rounds (including the original sender's round).

This is a geometric series with $a = 1$, $r = 5$, $n = 6$: $S_6 = \frac{1(5^6 - 1)}{5 - 1} = \frac{15624}{4} = 3906$.